



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering
Academic Year: 2025-2026(Odd Semester)

INNOVATIVE TEACHING PRACTICES

Degree, Semester & Branch: V Semester B.E. CSE

Course Code & Title: CCS375 Web Technologies

Name of the Faculty member: Dr. S. Poornam

Date & Time: 09.10.2025 & 10:30 A.M

Topic: Program Control

Course Outcome: CO4

Type of Activity: Interactive Learning Activity

Justification

This innovative classroom activity enhances student understanding of structured data representation and XML syntax correctness by engaging them in identifying and correcting errors in intentionally flawed XML documents. XML syntax errors such as missing closing tags, incorrect nesting, case sensitivity violations, and invalid attribute formats become real and practical challenges rather than theoretical concepts. This hands-on debugging approach fosters analytical thinking, problem-solving skills, and deep conceptual understanding of well-formed XML rules. By reconstructing valid XML from faulty code, students connect theoretical learning with practical application and real-time validation practices used in web development and data exchange systems.

Time Allotted for the Activity: 20 Minutes

Activity Preparation

- The instructor prepared printed sheets containing XML code samples with intentional errors related to:
 - ❖ Missing or mismatched closing tags
 - ❖ Case mismatches (e.g., <Student> vs <student>)
 - ❖ Incorrect or overlapping nested elements
 - ❖ Invalid or improperly formatted attributes
- Sample dataset sheets were also distributed.

Execution Steps

- ❖ Students were divided into groups of 4–5.
- ❖ Each group received a sheet containing incorrect XML code.

Tasks Assigned

- ❖ Identify all syntax and structure errors.
- ❖ Discuss and rewrite the XML as a valid, well-formed document.
- ❖ Final corrected XML was written on chart paper and displayed.
- ❖ Instructor verified the corrected document and clarified rules of well-formed XML.

Learning Outcome:

- ❖ Students clearly understood XML rules for well-formed documents.
- ❖ Students learned to identify structural inconsistencies and apply corrections logically.
- ❖ Improved understanding of nesting, tag pairing, and attribute formatting.
- ❖ Students gained confidence in working with XML as a data exchange format in web applications.

CO – PO / PSO Mapping

CO	PO1	PO2	PO3	PO4	PSO1
CO1	2	2	3	2	2

(1 – Low, 2 – Moderate, 3 – High)

Innovative practice	PO1	PO2	PO3	PO4	PSO1
	2	2	3	2	2
Justification for correlation	Students developed conceptual understanding of XML structure and syntax rules	Students analysed sample XML datasets to identify structural and formatting errors.	Students practiced debugging and reconstructing logically correct XML documents.	Students applied analytical reasoning to validate XML as real-time structured data.	Students applied XML validation skills for data representation and web-based applications.

PO/PSO mapped: PO1, PO2, PO3, PO4, PSO1

Images / Screenshot of the practice:



Reflective Critique:

- ❖ Students reported that the debugging challenge improved their understanding of XML formatting.
- ❖ They appreciated the practical approach instead of theory-only explanation.
- ❖ Students stated that identifying and correcting errors strengthened confidence when writing XML in future web-service projects.
- ❖ The competitive format increased engagement and motivation to achieve accuracy.
- ❖ Learners found it helpful to compare corrected versions across groups for deeper learning.

Benefits of the Practice:

- ❖ Converts abstract XML syntax rules into practical debugging experience.
- ❖ Encourages teamwork, communication, and collaborative problem solving.
- ❖ Strengthens observation and logical reasoning skills.
- ❖ Prepares students for real-time XML validation used in APIs, web services, and configuration files.
- ❖ Reinforces importance of accurate markup structure and hierarchical data organization.

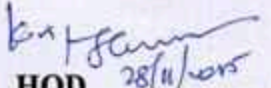
Challenges Faced in Implementation:

- ❖ Some students initially struggled with differentiating between logical and syntactic errors.
- ❖ Time management was challenging due to multiple errors requiring careful attention.
- ❖ Group coordination required guidance to maintain participation balance.

References

- ❖ <https://www.w3schools.com/xml/>
- ❖ <https://www.geeksforgeeks.org/xml-basics/>
- ❖ <https://www.tutorialspoint.com/xml/>
- ❖ <https://developer.mozilla.org/en-US/docs/Web/XML>


Signature of Faculty Member


HOD 28/11/25